

Rigorous Guide to Testing Independence

ORF 309

1 Fundamental Definitions

Definition 1 (Independence of Two Events). Events A and B are **independent**, denoted $A \perp B$, if and only if:

$$P(A \cap B) = P(A) \cdot P(B) \quad (1)$$

Definition 2 (Independence of Random Variables). Random variables X and Y are **independent**, denoted $X \perp Y$, if and only if:

Discrete case:

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y) \quad \text{for all } x, y \quad (2)$$

Continuous case:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y \quad (3)$$

CRITICAL DISTINCTION

Independent $\neq$ Mutually Exclusive

If $A \cap B = \emptyset$ and $P(A), P(B) > 0$, then A and B are **DEPENDENT**.

Why? Because:

$$\begin{aligned} P(A \cap B) &= 0 \quad (\text{mutually exclusive}) \\ P(A) \cdot P(B) &> 0 \quad (\text{both positive}) \\ \Rightarrow P(A \cap B) &\neq P(A) \cdot P(B) \end{aligned}$$

2 Equivalent Conditions for Independence

Theorem 1 (Equivalent Conditions for Independence of Events). If $P(A) > 0$ and $P(B) > 0$, the following are equivalent:

1. $P(A \cap B) = P(A) \cdot P(B)$ (**Definition**)
2. $P(A|B) = P(A)$ (**B doesn't affect A**)
3. $P(B|A) = P(B)$ (**A doesn't affect B**)
4. $P(A|B) = P(A|B^c)$ (**A 's probability unchanged by B**)
5. $P(B|A) = P(B|A^c)$ (**B 's probability unchanged by A**)

Proof Strategy

To prove independence: Show ANY one of these conditions holds.

To prove dependence: Show ANY one of these conditions FAILS.

Theorem 2 (Properties of Independence). *If $A \perp B$, then:*

- $A \perp B^c$
- $A^c \perp B$
- $A^c \perp B^c$

3 Step-by-Step Testing Procedure

3.1 For Events A and B

Step-by-Step Guide

Step 1: Identify what you know

- Do you know $P(A)$, $P(B)$, $P(A \cap B)$?
- Do you know conditional probabilities like $P(A|B)$ or $P(B|A)$?

Step 2: Choose your test

Option A: If you know $P(A)$, $P(B)$, and $P(A \cap B)$:

$$\text{Check if } P(A \cap B) \stackrel{?}{=} P(A) \cdot P(B) \quad (4)$$

Option B: If you know $P(A|B)$ and $P(A)$:

$$\text{Check if } P(A|B) \stackrel{?}{=} P(A) \quad (5)$$

Option C: If you know $P(A|B)$ and $P(A|B^c)$:

$$\text{Check if } P(A|B) \stackrel{?}{=} P(A|B^c) \quad (6)$$

Step 3: Make your conclusion

- If equal: Events are **independent**
- If not equal: Events are **dependent**

3.2 For Random Variables X and Y

Testing Independence of Random Variables

Method 1: Joint Distribution Factorization

Discrete: Check if $p_{X,Y}(x,y) = p_X(x) \cdot p_Y(y)$ for ALL (x,y)

Continuous: Check if $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$ for ALL (x,y)

Shortcut: If the joint PMF/PDF factors as $g(x) \cdot h(y)$ (one function of x only, one function of y only) over a rectangular domain, then $X \perp Y$.

Method 2: Conditional Distribution

Check if $P(X = x|Y = y) = P(X = x)$ for all x, y (discrete case)

Or equivalently: $f_{X|Y}(x|y) = f_X(x)$ for all x, y (continuous case)

Method 3: Expectation Test

If $X \perp Y$, then $E[g(X) \cdot h(Y)] = E[g(X)] \cdot E[h(Y)]$ for any functions g, h .

Special case: $E[XY] = E[X] \cdot E[Y]$

WARNING: The converse is FALSE. $E[XY] = E[X]E[Y]$ does NOT imply independence.

Method 4: Covariance Test

If $X \perp Y$, then $\text{Cov}(X, Y) = 0$ (uncorrelated).

WARNING: The converse is FALSE. $\text{Cov}(X, Y) = 0$ does NOT imply independence.

4 Worked Examples

Example 1 (Problem 1c from S2025 Midterm). **Setup:**

- $P(A) = 0.8$ (prob of getting question 1 correct)
- $P(B|A) = 0.6$ (prob of question 2 correct given question 1 correct)
- $P(B|A^c) = 0.1$ (prob of question 2 correct given question 1 wrong)

Question: Are A and B independent?

Solution:

First, calculate $P(B)$ using LOTP:

$$\begin{aligned} P(B) &= P(B|A)P(A) + P(B|A^c)P(A^c) \\ &= 0.6 \times 0.8 + 0.1 \times 0.2 \\ &= 0.48 + 0.02 = 0.5 \end{aligned}$$

Test 1: Compare $P(B|A)$ with $P(B)$

$$P(B|A) = 0.6 \neq 0.5 = P(B) \tag{7}$$

Conclusion: A and B are **DEPENDENT**.

Alternative Test: Compare $P(B|A)$ with $P(B|A^c)$

$$P(B|A) = 0.6 \neq 0.1 = P(B|A^c) \tag{8}$$

Same conclusion: **DEPENDENT**.

Verification using definition:

$$\begin{aligned}P(A \cap B) &= P(B|A)P(A) = 0.6 \times 0.8 = 0.48 \\P(A) \cdot P(B) &= 0.8 \times 0.5 = 0.4 \\P(A \cap B) &\neq P(A) \cdot P(B)\end{aligned}$$

Therefore **DEPENDENT**.

Example 2 (Geometric Wait Times - Problem 2 from S2025 Midterm). **Setup:** X = delay to New York ($\text{Geom}(0.1)$), Y = delay to Trenton ($\text{Geom}(0.2)$)

Given: The delays are probabilistically independent.

Question: Given $Y = 5$, what is $E[X|Y = 5]$?

Solution:

Since $X \perp Y$, conditioning on Y doesn't change the distribution of X :

$$E[X|Y = 5] = E[X] = \frac{1}{0.1} = 10 \quad (9)$$

Key insight: If $X \perp Y$, then any information about Y is irrelevant for X .

Example 3 (Testing with Joint PMF). **Setup:** Suppose (X, Y) has joint PMF:

$$p_{X,Y}(x, y) = \frac{1}{12} \quad \text{for } (x, y) \in \{(1, 1), (1, 2), (2, 1), (2, 2)\} \quad (10)$$

Wait, that doesn't sum to 1. Let me recalculate:

$$p_{X,Y}(x, y) = \frac{1}{4} \quad \text{for } (x, y) \in \{(1, 1), (1, 2), (2, 1), (2, 2)\} \quad (11)$$

Marginals:

$$\begin{aligned}p_X(1) &= p_{X,Y}(1, 1) + p_{X,Y}(1, 2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \\p_X(2) &= p_{X,Y}(2, 1) + p_{X,Y}(2, 2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \\p_Y(1) &= p_{X,Y}(1, 1) + p_{X,Y}(2, 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \\p_Y(2) &= p_{X,Y}(1, 2) + p_{X,Y}(2, 2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}\end{aligned}$$

Test independence:

$$\begin{aligned}p_{X,Y}(1, 1) &= \frac{1}{4} \\p_X(1) \cdot p_Y(1) &= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}\end{aligned}$$

Check for all pairs:

- $(1, 1)$: $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ ✓
- $(1, 2)$: $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ ✓
- $(2, 1)$: $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ ✓
- $(2, 2)$: $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ ✓

Conclusion: $X \perp Y$ (independent).

Example 4 (A Dependent Example). **Setup:** Roll a fair die. Let:

- A = event that die shows an even number
- B = event that die shows a number greater than 4

Calculate:

$$\begin{aligned}P(A) &= P(\{2, 4, 6\}) = \frac{3}{6} = \frac{1}{2} \\P(B) &= P(\{5, 6\}) = \frac{2}{6} = \frac{1}{3} \\P(A \cap B) &= P(\{6\}) = \frac{1}{6}\end{aligned}$$

Test:

$$\begin{aligned}P(A) \cdot P(B) &= \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} \\P(A \cap B) &= \frac{1}{6}\end{aligned}$$

Conclusion: $A \perp B$ (independent)!

This might be surprising, but it's correct.

5 Common Mistakes and Pitfalls

Common Mistakes

1. **Mutually exclusive $\neq$ independent**

If $A \cap B = \emptyset$ and $P(A), P(B) > 0$, then A and B are dependent.

2. **Uncorrelated $\neq$ independent**

$\text{Cov}(X, Y) = 0$ does NOT imply $X \perp Y$.

3. **$E[XY] = E[X]E[Y] \neq$ independence**

This is a necessary condition but NOT sufficient.

4. **$P(A|B) + P(A|B^c) \neq 1$**

The correct identity is: $P(A|B) + P(A^c|B) = 1$

5. **Pairwise independence $\neq$ mutual independence**

Events can be pairwise independent without being mutually independent.

6. **Must check ALL values**

For random variables, you must verify $p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y)$ for ALL (x, y) , not just some values.

7. **Given independence in the problem**

If the problem states events/variables are independent, USE IT! Don't try to verify it again.

6 Quick Reference Checklist

6.1 To Prove Independence

- ✓ Show $P(A \cap B) = P(A) \cdot P(B)$
- ✓ Show $P(A|B) = P(A)$ (or $P(B|A) = P(B)$)
- ✓ Show joint PMF/PDF factors: $p_{X,Y}(x, y) = g(x) \cdot h(y)$ over rectangular domain
- ✓ Use given information (if problem states they're independent)

6.2 To Prove Dependence

- ✓ Show $P(A \cap B) \neq P(A) \cdot P(B)$
- ✓ Show $P(A|B) \neq P(A)$ (or $P(B|A) \neq P(B)$)
- ✓ Show $P(A|B) \neq P(A|B^c)$ (or $P(B|A) \neq P(B|A^c)$)
- ✓ Show joint PMF/PDF does NOT factor for some (x, y)

6.3 Warning Signs of Dependence

- Problem says "given that..." and it affects probabilities
- Conditional probability differs from marginal probability
- Events describe related outcomes (e.g., "both depend on understanding")
- Joint distribution doesn't factor

7 Decision Tree

